

1. GENERATE

initial generation

guided generation prompt

Given the initial user prompt $\{\text{user_prompt}\}$, best answers $\{\text{best_group}\}$ and the improvement recommendations $\{\text{gradient}\}$, generate an improved answer

LLM

h_1
 h_j
...
 h_N

generate answers $\{h\}$ based on the user query and textual gradient (in subsequent stages)

2. JUDGE



judge prompt P_j

...

+

... Does this answer $\{h_j\}$ meet the criterias $\{\text{criteria}_1\}, \dots, \{\text{criteria}_K\}$?

judge response

The response is ...
therefore the response is $\langle \text{criteria}_1 \rangle$
 $\langle \text{criteria}_1 \rangle$
...
 $\langle \text{criteria}_K \rangle$
 $\langle \text{criteria}_K \rangle$

$P(\text{label}_1|P_j)$

yes
good
correct
false
no
...

$P(\text{label}_K|P_j)$

helpful
friendly
useful
useless
harmful
...

hard labels

hard label l_1

...

hard label l_K

contrastive margins

$s_1(h_j) = ((\sum \log P(\text{pos})) / |\Pi_1| - (\sum \log P(\text{neg})) / |\Sigma_1|)$

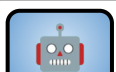
...

$s_K(h_j) = ((\sum \log P(\text{pos})) / |\Pi_K| - (\sum \log P(\text{neg})) / |\Sigma_K|)$

3. REFLECT

Synthesize gradient

gradient
Here are the recommendations on how to improve an answer...



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You are a preference gradient calculator.
...
chosen group: $\{G\}$
rejected group: $\{B\}$...

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Rank

$\left\{ \begin{array}{l} h_1, \{l_{11}, \dots, l_{1K}\}, \sum_{k=1..K} w_k * s_k(h_1) \\ \dots \\ h_N, \{l_{N1}, \dots, l_{NK}\}, \sum_{k=1..K} w_k * s_k(h_N) \end{array} \right\}$

Contrastive groups

good group

$G = \{h_{i1}, h_{i2}, \dots, h_{i|G|}\}$

bad group

$B = \{h_{i1}, h_{i2}, \dots, h_{i|G|}\}$

$\{h_q\} | h_q \notin G \cup B$

have we reached the maximum number of iterations?



return h_{best}

User prompt